
GramVest

SIH 2026 • PS 26091

M3 - GIS, Market Intelligence, Competitor Mapping & Opportunity

Module implementation handbook - GIS / market owner

TEAM IMPLEMENTATION HANDBOOK

Prepared as a practical engineering guide for independent module development, stable integration, testing, and handoff.

Source basis: SIH 26091 Implementation Blueprint and the corresponding GramVest module
Markdown handbook.

Contents

1 1. Mission	1
2 2. Why this module matters	1
3 3. Responsibilities	1
4 4. Recommended tools	1
5 5. Repository	2
6 6. Location workflow	2
7 7. PostGIS data model	2
8 8. Radius analysis	3
9 9. Market Reach	3
10 10. Customer estimation — keep assumptions explicit	3
11 11. Competitor Mapping	4
11.1 List data	4
11.2 Map data	4
12 12. Competitor density	4
13 13. Category saturation	4
14 14. Market access	5
15 15. Underserved niche detection	5
16 16. Opportunity Engine	5
17 17. Pricing / Market Value	6
18 18. Data quality rules	6
19 19. Demo dataset workflow	6
19.1 Step 1	6
19.2 Step 2	6
19.3 Step 3	6
19.4 Step 4	6
19.5 Step 5	6
19.6 Step 6	7
19.7 Step 7	7
19.8 Step 8	7
19.9 Step 9	7
20 20. Output contract for M2/M1	7
21 21. Integration with M4	7
22 22. Integration with M5	7
23 23. Definition of done	7

1. Mission

Build the hyper-local intelligence layer that answers:

- Where is the business?
- Who is nearby?
- How large is the reachable market?
- How much competition exists?
- Where are gaps or underserved areas?
- Why does a business appear attractive or unattractive in this location?

The blueprint identifies **local context as the USP** and calls for a configurable 5 km/10 km analysis radius, market reach, competitor density, market access, saturation, underserved niches, and evidence/confidence.

2. Why this module matters

A generic business advisor can say “dairy is a good business.” GramVest must say something more useful:

“In this specific location, there are approximately X potential customers, Y relevant competitors, and an underserved area in Z direction.”

That location-aware reasoning is what differentiates the product.

3. Responsibilities

M3 owns:

1. geocoding/location resolution
2. analysis radius
3. demographic/household inputs
4. market reach estimates
5. nearby market/town/transport data where available
6. business/competitor dataset normalization
7. PostGIS spatial queries
8. competitor density calculations
9. category saturation
10. underserved-area identification
11. market evidence + confidence metadata
12. opportunity inputs and evidence
13. market-related data quality tests

4. Recommended tools

Task	Tool
GIS database	PostgreSQL + PostGIS
Map prototype	Leaflet + OpenStreetMap
Geocoding	agreed geocoding provider or dataset; keep provider abstracted
Data cleaning	pandas
Analysis	pandas / NumPy
Visualization/debug	QGIS (optional but very useful)
Database inspection	DBeaver/pgAdmin
API testing	Postman
Version control	Git + GitHub

The blueprint recommends PostGIS for radius/distance queries and Leaflet + OpenStreetMap for the prototype map.

5. Repository

```
data/  
├─ raw/  
│   ├── geography/  
│   ├── businesses/  
│   ├── markets/  
│   ├── demographics/  
│   └── prices/  
└─ processed/  
    ├── businesses/  
    ├── markets/  
    ├── demographics/  
    └── prices/
```

```
backend/services/  
├─ geo/  
├─ market/  
├─ competitor/  
└─ opportunity/
```

6. Location workflow

Input:

```
village/town  
block  
  district  
state  
optional map pin
```

Flow:

```
User location text  
→ normalize  
→ geocode/resolve  
→ latitude/longitude  
→ district/block validation  
→ choose 5 km or 10 km radius  
→ store location context
```

Output:

```
{  
  "village": "Example Village",  
  "block": "Example Block",  
  "district": "Jalandhar",  
  "state": "Punjab",  
  "latitude": 31.325,  
  "longitude": 75.576,  
  "analysis_radius_km": 5,  
  "confidence": 0.92  
}
```

These values are example structure only.

7. PostGIS data model

Businesses table should include:

id
name
category
subcategory
latitude
longitude
geom/geography
source
source_date
confidence

For PostgreSQL/PostGIS, store a spatial point suitable for distance queries.

8. Radius analysis

Primary presets:

- 5 km
- 10 km

The blueprint explicitly recommends these presets.

Conceptual query:

```
SELECT *  
FROM businesses  
WHERE ST_DWithin(  
    businesses.geom,  
    user_point,  
    radius_meters  
);
```

Use proper geography/geometry types consistently according to the chosen schema.

9. Market Reach

Inputs may include:

- population
- households
- geographic radius
- business category
- accessibility
- local economic signals

Outputs:

population
households
estimated_addressable_customer_segment
market_access

The blueprint requires evidence/confidence for every major estimate.

10. Customer estimation — keep assumptions explicit

Do NOT claim exact customer counts unless supported by the dataset.

A simple prototype model may be:

```
estimated_addressable_customers  
= households × target_customer_rate × accessibility_factor
```

But the actual `target_customer_rate` and `accessibility_factor` must be stored as assumptions.

Return:

```
{
  "potential_customers": 1210,
  "confidence": 0.78,
  "assumptions": [
    "target customer share = ...",
    "5 km radius",
    "category = dairy"
  ]
}
```

Do not hide model assumptions.

11. Competitor Mapping

M3 should return both:

List data

business_name
category
distance_km
estimated_scale/source field if available
source
source_date
confidence

Map data

```
{
  "id": "c1",
  "lat": 31.32,
  "lng": 75.57,
  "name": "Example Dairy",
  "category": "dairy",
  "distance_km": 1.8
}
```

12. Competitor density

Possible simple metric:

`competitor_count / analysis_area`

But present to the user as:

- low
- moderate
- high

with explanation.

Do not claim “no competition” simply because no businesses were found in a dataset.

13. Category saturation

Compare relevant businesses against estimated target customer base or a configured local capacity heuristic.

Keep the model explainable.

Return:

saturation_score
interpretation
confidence

14. Market access

Where data is available, use:

- distance to market/town
- transport accessibility
- travel time
- road connection indicators

Do not introduce unsupported precision.

15. Underserved niche detection

Look for:

- clusters of customers with low relevant competitor coverage
- locations far from category providers
- category mismatch
- underserved price bands where data exists

Output an evidence list rather than only a score.

Example:

Opportunity zone:
Eastern residential corridor

Why:

- adequate household density
- fewer relevant businesses
- acceptable access

Confidence:

0.72

16. Opportunity Engine

The blueprint says opportunity should combine:

- market size
- competition
- affordability
- business fit
- local gaps

The recommendation must explain why the business is attractive.

Suggested result:

```
{  
  "score": 76,  
  "market_score": 82,  
  "competition_score": 68,  
  "affordability_score": 73,  
  "business_fit_score": 78,  
  "gap_score": 81,  
  "summary": "...",  
}
```

```
"evidence": [],  
"confidence": 0.8  
}
```

17. Pricing / Market Value

If price data is available:

Store:

```
product  
unit  
location  
price  
date  
source
```

Return:

- indicative price range
- median/reference price if available
- date
- source
- confidence

Never present old/stale data as live.

18. Data quality rules

Test:

- duplicate businesses
- impossible coordinates
- missing category
- stale dates
- inconsistent units
- duplicate market locations
- outlier prices
- missing households

The blueprint explicitly calls for coordinate, duplicate, stale-price and missing-value tests.

19. Demo dataset workflow

Step 1

Pick one demo geography.

Step 2

Pick 2–3 business categories.

Step 3

Collect/prepare business records.

Step 4

Normalize categories.

Step 5

Validate coordinates.

Step 6

Load into PostgreSQL/PostGIS.

Step 7

Run 5 km/10 km queries.

Step 8

Manually inspect maps in QGIS or the frontend.

Step 9

Freeze demo data version.

20. Output contract for M2/M1

Provide:

LocationAnalysis
MarketSummary
CompetitorSummary
OpportunityEvidence
PriceSignals

Each should contain confidence + source metadata.

21. Integration with M4

M3 supplies:

- market score
- competition score
- customer estimates
- affordability signals
- opportunity evidence

M4 uses these as inputs to viability scoring.

22. Integration with M5

M3 supplies structured evidence for:

- recommendation explanations
- SWOT
- risk
- AI answers

M5 must never invent market numbers when M3 does not provide them.

23. Definition of done

M3 is done when:

- location resolution works for demo area
- 5/10 km queries work
- competitor map works
- market summary works
- opportunity evidence works
- confidence/source metadata exists

- data quality checks exist
- API response matches M2 contract
- at least one complete demo business works end-to-end